AI-Assisted Single-Cell Diagnostic Prototype

Multimodal Optical Microscopy & Electrochemical Sensor Analysis

CAUTION — RESEARCH PROTOTYPE: *RESEARCH PROTOTYPE NOTICE: This system is developed strictly for research, educational, and computational evaluation purposes. It is not an FDA-cleared medical device and must not be used as a clinical diagnostic tool or to guide medical treatments.*

Sample ID:	Test_Abnormal_Cell	**Analysis Date:**	2026-10-22 18:22:46
Specimen Type:	Single Erythrocyte (RBC)	**Multimodal Modalities:**	Optical Brightfield + CV / EIS

Diagnostic Classification Findings

Multimodal Prediction	Confidence	Research Risk Score
ABNORMAL	**100.0%**	**1.00 / 1.00**

Multimodal Biomarker Quantitative Extraction

Image Feature	Value
Cell Area	2793.0
Circularity (0-1)	0.1982
Aspect Ratio	0.5391
Texture Variance	1572.6
Mean Intensity	7.1

Electrochemical Feature	Value
Peak Anodic Current (Ipa)	31.58 µA
Charge Transfer Res. (Rct)	568.03 Ω
Capacitance (Cdl)	4.77 µF
Peak Potential Diff (ΔEp)	111.56 mV
Impedance |Z| at 1 Hz	614.54 Ω